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Exploring the influence of audience familiarity on speaker anxiety and performance in virtual reality and real-life presentation contexts

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ABSTRACT

Virtual reality (VR) offers immense freedom in the design of virtual instructional environments, but little guidance exists on how to capitalise on this freedom. This article reports on a study exploring how audience familiarity influences public speaking anxiety (PSA) and performance in a presentation speaking task *in virtuo* and *in situ*. Questionnaire instruments were used to gauge the PSA, motivation, focus, and self-confidence of 10 undergraduate students who each presented in four different audience conditions across VR and real life. Presentations were transcribed to identify features of performance, including utterance fluency, and speaking breadth and depth. Outcomes indicated that an audience of computer-generated agents resulted in less PSA than an audience of photorealistic people familiar to the speakers. Additionally, presenting to an audience of strangers in real life induced the most anxiety, but the performance features of articulation rate, disfluencies, and frequency of silent pauses were significantly improved in this condition. The main contribution of this study is to show that presentations directed at virtual audiences exhibit less fluent speech in non-native speakers than speeches to a real audience.

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Virtual reality; public speaking anxiety; speaking performance; audience familiarity

1. Introduction

Virtual Reality (VR) technology has made itself increasingly difficult to ignore in the fields of education and training. One of the advantages that VR brings to practicing presentations is the ability to tailor the presentation conditions for optimal performance, even at the individual student level. VR presentation simulations offer an alternative to practicing and delivering presentations in reality, thereby allowing users to observe, interact, and behave largely uninhibited by consequences which may occur performing similar actions *in situ*. Trainers have taken advantage of this affordance, using VR to train pilots and surgeons who have high stakes jobs, where the consequences of a mistake can be serious (Sherman and Craig 2019). Additionally, VR has been used to help people alleviate phobias and anxiety by putting users into safe and controlled simulations to help them deal with their fears (Carlin, Hoffman, and Weghorst 1997; Sharmili and Kanagaraj 2020), including the fear of public speaking (Harris, Kemmerling, and North 2002; Nazligul et al. 2018; Slater et al. 2006).

The safe and controlled nature of virtual environments has been useful in language learning classrooms

as well (Lan 2020). Through practicing in VR, speakers of an additional language can mitigate their feelings of nervousness, inhibition, anxiety, and fear of losing face that may arise when using a learned language to give a presentation or speech (Lin and Lan 2015; Schwienhorst 2002). VR-based presentation training applications are becoming more common (e.g. *Ovation VR*,¹ *VirtualSpeech*²), and offer linguistic and metalinguistic feedback to users in addition to helping lower anxiety (Stupar-Rutenfrans, Ketelaars, and van Gisbergen 2017). However, the familiar pattern of commercial technology outpacing the research is particularly acute with VR presentations. Most research in this stream has focused on user performance or anxiety from the effects of virtual audience behaviour (Boetje and van Ginkel 2021; Chollet and Scherer 2017; Kang et al. 2014; Kang et al. 2016), or on the presence, absence, or size of a virtual audience (Mostajeran et al. 2020; Slater et al. 2006), but no known studies have examined differences in audience member appearance. Therefore, the research presented here brings necessary attention to how variations in audience member appearance, namely the familiarity of their faces, in VR and *in situ*

contexts may affect anxiety and performance in a language learning context.

2. Literature review

Performance in an oral presentation can be considered part of a triadic reciprocal causal framework which also includes the instructional environment and individual speaker characteristics (Bandura 1997; Boetje and van Ginkel 2021). These three factors (speaker characteristics, such as trait anxiety, or other affective or cognitive factors; the environment where the presentation takes place, either virtual or real; and the student's performance during the presentation) are all equivalent interacting determinants. Within this framework, it is possible to manipulate the instructional environment, for example, in order to observe reactions in student performance in an oral presentation, with the goal of optimising the instructional design.

2.1. The virtual instructional environment

VR affords a sense of presence in users, which is described as a combination of perceiving one's self as being in a place, and that occurrences in that place have consequences (Slater 2009). A sense of presence in a virtual environment has been shown to cause individuals to behave as they would in a similar situation in real life (RL) (Slater and Sanches-Vives 2016). This has been observed in public speaking simulations, where participants who were predisposed to public speaking anxiety (PSA) showed higher levels of anxiety while presenting in front of a virtual audience as opposed to an empty room in VR (Slater et al. 2006).

There is no limit to the manipulations that can be made to a VR instructional environment. This provides the freedom to tailor these environments to the individual in order to enhance performance. For instance, many studies of VR presentations examine speaker perception of the virtual audience, primarily focusing on virtual audience behaviour as communicated through programmed body language showing arousal (alertness) and valance (positive or negative feelings towards the speaker) (Chollet and Scherer 2017; Glémarec et al. 2021; Kang et al. 2014). These studies have shown that specifically programmed body language, such as leaning, crossing arms, and gaze direction, can influence speaker anxiety, as they perceive these actions to reflect states of arousal or valance (Chollet and Scherer 2017; Kang et al. 2014).

Other studies have also examined audience behaviour. For example, in a qualitative inquiry on the

effect of virtual audiences on user experience, Gruber and Kaplan-Rakowski (2020) reported an array of reactions from students presenting in a foreign language in VR. Their virtual audience consisted of student-like virtual agents in a classroom setting which were programmed with body language such as nodding or shifting the torso from side to side. Student presenters ($n = 12$) reported varied perceptions of the audience, contrasting between seemingly real to obviously artificial. Regardless, students behaved in front of the audience during their VR presentations as they would have in front of a real audience by, for example, using common strategies such as making eye contact with audience members or avoiding eye contact by looking past the audience to a point in the back of the virtual venue.

The size of the virtual audience also appears to impact PSA. Comparing VR and RL presentation environments, a study by Davis et al. (2019) evaluated PSA levels between native speakers who presented in front of a large virtual audience to those who presented in front of a single peer in RL. They found that PSA was much higher in participants who presented to the large VR audience, although no difference was found between the groups' presentation performance. A study by Mostajeran et al. (2020) also looked into social anxiety in public speaking contexts with audiences of three, six, and fifteen virtual humans. Contrary to expectations, they reported that virtual audiences of three produced more stress and higher heart rates than audiences of six or fifteen.

In looking at the fidelity of the virtual instructional environment, Dubiago, Poeschl, and Doering (2017) examined how presenters were affected by a combination of scene realism and audience behaviour. They found that these two factors significantly predicted the overall perception of the simulation realism. Wilsdon and Fullwood (2017), in a similar study, measured the effects of increased simulation fidelity on PSA and found no changes in PSA as fidelity increased. So, it appears that although presenters are sensitive to the fidelity of the simulation, it may not influence PSA levels. However, Dubiago, Poeschl, and Doering (2017) speculated that whereas their audience appeared more casual, a more formal looking audience may have a different effect on presenters, and suggested that future studies examine the effect of audience appearance on presenters. With the advent of more sophisticated and assessable 3D scanning techniques, one such manipulation of appearance of a virtual audience could be audience familiarity, wherein virtual agents with appearances familiar to the presenter are used as audience members.

Audience familiarity has been shown to impact PSA in RL contexts, wherein more familiar audiences usually caused less anxiety (MacIntyre and Thivierge 1995). Familiarity can be perceived in degrees, from complete strangers through acquaintances and on to close friends and family. Considering appearance only, adults can become familiar, to a degree, with novel faces after only one minute of exposure, recognising them again even when expressions and hair styles are varied from the initial exposure (Baker, Laurence, and Mondloch 2017). Consequently, it seems likely that an individual who is introduced to someone new, even briefly so, may feel acquainted with them upon a second encounter. Familiarity can be manipulated in virtual instructional environments by introducing virtual agents who are acquainted with the presenter, with the intention of mitigating PSA. Although the effect of familiarity is understood in RL presentation contexts, whether its effect extends to VR contexts, as audience behaviour and size have been shown to, is unknown.

2.2. Individual characteristics

The idiosyncrasies that speakers bring to the learning environment may be neglected due to the limited amount of time and attention educators can spend on each student. But technology such as VR can help mitigate these constraints through the flexibility of tailoring the virtual instructional environment. For example, speakers may be influenced by PSA under different circumstances (Tudor, Poeschl, and Doering 2013), and VR can be used to moderate or alleviate this through exposure therapy or by adjusting those elements which individuals react to with anxiety (Harris, Kemmerling, and North 2002; Heuett and Heuett 2011; Lister, Piercey, and Joordens 2010; Stupar-Rutenfrans, Ketelaars, and van Gisbergen 2017; Takac et al. 2019). The dynamic nature of anxiety requires consideration of a variety of factors which may influence it, such as individual characteristics, as emphasised by MacIntyre (2017).

Zhou et al. (2021), for example, examined the role of gender, personal confidence, and self-bias alongside PSA in a VR presentation training task. Although gender showed no significant effect, PSA, personal confidence, and self-bias were seen to correlate significantly, suggesting that individuals with lower confidence may improve their self-assessment and reduce anxiety after practicing presentations in VR. Considering the impact that individual differences may have in terms of PSA and task performance, the following characteristics were included in this study.

2.2.1. Motivation

VR has been identified as a motivating learning technology through its unique affordances of immersion and presence (Dalgarno and Lee 2010). It has also been noted by Pinel and Csizér (2015) that motivation is significantly related to anxiety, wherein highly motivated individuals were found to have lower levels of anxiety. Higher levels of motivation may in fact decrease anxiety, and may also lead to greater achievement in language learning (Yan and Horwitz 2008). As both motivation and anxiety are dynamic, i.e. subject to influence from contextual factors across time, it was necessary to account for the influence of VR as a motivating factor in the present study to better understand the nature of reported PSA.

2.2.2. Self-confidence

MacIntyre (2017) outlined a number of effects of language anxiety, which included reduced linguistic self-confidence. Self-confidence has been shown to be a significant contributor to willingness to communicate in an additional language (Ghanbarpour 2016), and a lack of self-confidence has been found to be a common reason for PSA (Raja 2017). Furthermore, several researchers have found a significant link between self-confidence and proficiency in an additional language (Clément, Dörnyei, and Noels 1994; Matsuda and Gobel 2004; Menzel and Carrell 1994). On the other hand, less is known about whether self-confidence is affected by virtual environments, and given its association with both anxiety and speaking performance, self-confidence is accounted for in the present study.

2.2.3. Focus

Last, one of the primary cognitive effects of anxiety is increased self-related cognition, including thoughts of failure and self-deprecation (MacIntyre 2017). Individuals with high anxiety have been shown to pay less attention to their environment and have more negative inwardly focused thoughts about their performance, which in turn may lead to poorer speaking performance (Daly, Vangelisti, and Lawrence 1989). Whereas the effects of virtual audiences on PSA have been examined, the behaviour of self-focused attention in VR scenarios is not fully understood (Bouchard et al. 2014). It is possible that VR presentation simulations may help decrease self-focused attention, allowing for task-focused attention, and therefore the degree to which a VR presentation simulation affects task-focused attention is included in the present study.

2.3. Task performance

2.3.1. Public speaking anxiety

As mentioned, PSA can be an influential factor in presentation performance. It is often considered an outcome of environmental and psychological conditions which can then impact task performance (Zhou et al. 2021). General anxiety is an affective phenomenon described as consisting of feelings of tension, apprehension, and worry, which can be triggered by a specific event (state anxiety) or exist as a general behavioural disposition (trait anxiety) (Spielberger 1966). PSA is a common and often debilitating state anxiety (Menzel and Carrell 1994; Woodrow 2006) and can be reliably measured with survey instruments such as the Personal Report of Confidence as a Speaker (PRCS; Gilkinson 1942).

2.3.2. Spoken output

Performance in a presentation speaking task can be measured in a variety of ways (Bankowski 2010), but objective, task independent measures, such as spoken output, can lend themselves readily in presentation research. Utterance fluency, for example, is a key criterion by which examiners and language assessors gauge speaking ability, and it is also known to be impacted by PSA (Castillejo 2018; Woodrow 2006). Segalowitz (2010) defines utterance fluency as the objective speed, repair, and breakdown characteristics of the delivered language, as measured by articulation rate and disfluencies such as repetitions and repairs. Non-strategic hesitation markers, such as infelicitous silent pauses during a presentation, can also be used as a measure of fluency (Castillejo 2018). Moreover, both Barreda-Ángeles, Aleix-Guillaume, and Pareda-Baños (2020) and Wörtwein, Morency, and Scherer (2015) have observed a higher frequency in silent pauses to associate with PSA, making silent pauses a potential sign of anxiety as well. Environmental and individual factors can affect cognitive processes and in turn lead to changes in utterance fluency. For example, MacIntyre (2017) noted that anxiety is known to disrupt the cognitive retrieval of information and consequently influences utterance fluency. Therefore, observed changes in utterance fluency under conditions where anxiety is detected may be indicative of anxiety being the underlying cause of those changes.

In addition to fluency measures, breadth and depth of a presentation can be measured in terms of total speaking time and pruned word count. Presenters who spoke for longer and who contributed a larger number of lexical items would indicate higher processing fluency, which is related to utterance fluency (De Jong

2018; Segalowitz 2010), and demonstrates a willingness to endure the presentation condition, a potential indicator for the presence or absence of PSA.

2.4. Research questions

The above review of research indicates that due to a sense of presence afforded by VR, individuals are affected similarly when giving a speech or presentation in front of a virtual audience as when they do so in front of an audience in RL in terms of audience size and behaviour. It is also evident that these variations in audience behaviour and size can influence PSA. There is additional evidence that PSA negatively affects the fluency of foreign language speakers in RL contexts. However, it is not yet known if PSA from presenting to VR audience has the same effect on fluency or other performance features. Nor is it known if the familiarity of the audience affects the level of PSA similarly to the way audience behaviour and size do. Therefore, the present research is guided by the following questions:

RQ1: How do variations in audience familiarity associate with reported PSA in RL and VR?

RQ2: Do self-confidence, motivation, or focus correlate with PSA in VR conditions?

RQ3: How do variations in audience familiarity associate with speaking performance?

3. Method

To answer these research questions, four experimental conditions were created for a quasi-experimental, within-subjects study design. The first two conditions were ‘virtual reality familiar’ (VR-F) and ‘real-life familiar’ (RL-F), which both used the same audience resemblance through photorealistic scanned 3D images of the heads of the researchers in VR, who were also seated corresponding to their virtual counterparts in the RL condition. As the researchers met with each of the participants in the initial stages of the study, they were considered to be acquainted or mildly familiar with them. The second pair of conditions were virtual reality unfamiliar (VR-U) and real-life unfamiliar (RL-U). The VR-U condition used computer-generated virtual agents for audience members while the RL-U condition audience was composed of human individuals not previously encountered by the participants. Participants were randomly assigned to first present to either the familiar or unfamiliar conditions, and then switched to engage with the other condition. Figure 1 depicts the two VR conditions and Figure 2 depicts a participant presenting



Figure 1. VR-F audience condition (left) and VR-U audience condition (right).

in VR in the same environment used for the RL condition.

3.1. Participants

Eleven participants were recruited for participation in the study on a non-random, volunteer basis. One participant abandoned the study (P8), leaving 10 participants who completed it (6 females, $M_{age} = 19.1$). All participants were Chinese undergraduate students in their first or second year at a Sino-British university in mainland China, and according to university placement exams, were at least at the B1 level in their use of English in regards to the Common European Framework of Reference (CEFR; Council of Europe 2001). Participants declared themselves to be willing and

able to complete the study, with no anxiety or speech disorders.

3.2. Presentation environments and apparatus

A custom-made VR presentation simulator was created for this study, which renders a 3D virtual environment of an audience of four in a small office. Four virtual agents acquired from the Unity asset store comprised the virtual audience in the VR-U condition. In the VR-F condition, the heads of these agents were replaced with the heads of the researchers. The researchers' heads were scanned using Microsoft's Kinect 3D scanner application running on Windows 10. The participants embodied an avatar facing the audience that was also acquired from the Unity asset store. Neither the avatar



Figure 2. Participant presenting using VR (showing classroom used for the RL environment).

nor controllers were visible to the participants in the virtual environment. Participants could see a text box in their lower field of view which contained a notes section with bullet points for each slide of the presentation. The slides were displayed on the wall behind the speaker's avatar and could be viewed if the speaker looked over their shoulder. The RL presentation environment was nearly identical to the virtual environment in room-size, furniture layout, audience size and slide projection placement.

The virtual agent audience were animated with body language such as gaze direction, arm-crossing and leaning toward or away from the presenter to convey arousal and valence and support a sense of social presence (Oh, Bailenson, and Welch 2018). Animations were programmed to avoid movements occurring in unison to make the virtual audience appear natural. The audience remained idle until the presenter pressed a start button on their controller which caused the audience to clap for about 4 s. The participant would also end the presentation by pressing a button on the controller which would again produce the clapping animation. Participants were not informed whether the virtual agents were programmed or puppeteered.

An Oculus Rift + Touch was used to access the virtual environment, which includes a head mounted display and two handheld controllers. The Touch controllers were used for both the VR and RL presentations, functioning to transition between slides. The Oculus Rift headset has a resolution of 1280 × 800 and was connected with cables to a laptop powered by an i7 and had 16GB of RAM and a GeForce RTX 2070 GPU. There were no noticeable latency issues. We used a standard 27" monitor facing the participant during RL presentations to display the notes section. Text size was calculated to be the same in VR and RL.

3.3. Procedure

Two presentations were designed by professional English language teachers to be at a level appropriate for the participants; one for the familiar audience conditions and one for the unfamiliar audience conditions. Both presentations were on unfamiliar topics to minimise the chances of an artificially better performance by some or all participants (Kazemi and Zarei 2015). The two topics were the lifestyle of Neanderthals and interstellar travel. The presentations were formatted as a slide show similar to Microsoft PowerPoint. The slides contained short titles and between one to three images. Participants gave each presentation twice, once in VR and once in RL, with a minimum time between presentations of two days. The experiment took place over two

weeks, during which participants were required to attend four sessions, one for each condition. Participants were allowed up to five minutes to deliver their presentation, but there was no minimum time limit.

A short tutorial of the VR equipment was provided to familiarise participants with the technology in order to reduce cognitive load which can affect performance and anxiety (Chen 2020; King and Finn 2017). For their first session with a topic, they were given 10 min to practice the presentation, and for their second encounter with the topic they were given five minutes to refresh their memories. For practice sessions, participants had a full speech script in the notes section with an average word count of 305 words per presentation. The actual presentation had the script replaced by bullet points which contained sentence abbreviations from the full presentation with an average word count of 80.5 words per presentation. Practice sessions were not sufficiently long to memorise the presentations, and with minimal repetitions and the inclusion of novel words, the effect of repetition on fluency was likely minimal (Sun and Révész 2021; Yoshimura and MacWhinney 2007). Researchers left the room during all practice sessions and while participants were giving presentations in VR to avoid the sense of being watched, which could affect the sense of immersion and anxiety (Fernandes et al. 2016). Participants were made aware that presentations were voluntary and not graded.

3.4. Survey instruments and reliability

A battery of three short surveys (all paper and pen, in English) was used to measure PSA, motivation, self-confidence, and focus. The surveys were administered at intervals over the course of the study as follows.

- (1) A custom-made self-confidence survey was administered before participants began. It contained five items to measure demographic information and three correlated Likert-style items (from disagree to agree) to measure self-confidence in giving presentations in English. These items were (a) *I feel confident when I speak English in front of others*; (b) *I am shy when I have to give presentations in English*; and (c) *I feel positive about my ability to give presentations in English*.
- (2) The anxiety survey was administered directly after participants delivered each presentation, for a total of four measurements. The survey was composed of the short-form of the Personal Report of Confidence as a Speaker (PRCS) from Hook, Smith, and Valentiner (2008), designed to measure

PSA, and which contains 12 items. Items were in a mixed order at each delivery of the survey.

- (3) A custom-made reflective survey was administered after participants gave their fourth and final presentation of the study. This included three correlated Likert-style items to measure motivation from VR use, as follows (a) *I would like to continue using this VR program to practice my presentations*; (b) *The VR program makes me want to get better at giving presentations*; and (c) *The VR program motivates me to practice giving presentations*; and three correlated Likert-style items to measure the ability to focus on the presentation task in VR, which were (a) *The VR program helped me to be more focused when giving presentations*; (b) *I think the VR program was distracting when giving presentations*; and (c) *It was easy to focus on presenting while using the VR program*.

The constructs measured with the survey battery showed an adequate reliability (see Table 2, section 4) with an alpha coefficient above 0.7 for each construct (Dörnyei and Taguchi 2010).

3.5. Spoken output data

In addition to survey data collected, each presentation was video recorded, with participants' permission, and transcribed manually. A total of 174.6 min of recorded spoken output was collected for transcription.

Two indicators of fluency were used, adapted from De Jong (2018), which were articulation rate and frequency of errors (disfluencies). The fluency measures used, and how they were calculated, is included in Table 1.

In regards to disfluencies, repetitions were counted when occurrences of a word, partial word, or phrase were found in succession and the repetition was determined not to serve a communicative function. Repairs were identified as either a reformulation of speech (when a word or phrase is communicatively synonymous but syntactically different with a word or phrase in succession to it), or as a false start (when an utterance is abandoned before completion and the speaker starts a

new, different utterance). The excerpt in Figure 3 shows the disfluencies measured.

Three additional measures of spoken performance included in the analysis of the transcript data were total speaking time (TST), pruned word count, and mean frequency of silent pauses. TST (total time minus silent pauses) was analysed independently to determine breadth of output. Pruned word count was also reported, which is the total word count minus any disfluencies. For example, line 55 in Figure 2 would have a pruned word count of three. The pruned word count was used to evaluate depth of linguistic output when accounting for the words supplied in the bullet-point notes in each presentation. Lastly, mean frequency of non-strategic silent pauses was included as a potential indicator of anxiety but not fluency because the nature of the presentation task which involved participants scanning bullet-point notes and looking at images on slides corrupted the reliability of this measure for measuring fluency.

3.6. Data analysis

To answer RQ1, comparisons were made between audience conditions (familiar and unfamiliar) and delivery modes (VR or RL) using a Wilcoxon Signed Rank Test, which is suggested for non-parametric data (Field 2009). For RQ2, a Kendall's Tau correlation, which is recommended for small samples (Field 2009), was conducted to compare variables. And for RQ3, spoken output was analysed via transcripts of each presentation to identify performance features, and relevant comparisons were made using a Wilcoxon Signed Rank Test.

4. Results

The results of the survey battery can be seen in Table 2. The anxiety scores are reported as a percentage of total possible anxiety from the short-form PRCS. For RQ1, there was a nonsignificant difference in PSA levels between participants for VR-F and RL-F audiences ($p = .4$). However, participants had significantly higher anxiety levels in the RL-U condition ($Mdn = 2.59$) than in the VR-U condition ($Mdn = 2.04$), $z = -2.1$, $p < .05$, $r = -.47$. Furthermore, between VR conditions, there was significantly more anxiety with the familiar audience than the unfamiliar audience, $z = -2.197$, $p < .05$, $r = -.5$. Between RL conditions, differences in anxiety were nonsignificant between familiar and unfamiliar audiences ($p = .33$).

Regarding RQ2, participants' confidence in delivering a presentation in English was slightly more than neutral on average, although, both motivation from

Table 1. Fluency measures used in the present study.

Measure	Formula
Articulation rate (per second)	Number of syllables divided by total speaking time
Number of repetitions (per minute)	Number of repetitions divided by total speaking time
Number of repairs (per minute)	Number of repairs divided by total speaking time

Table 2. Alpha coefficient (α), percent PSA, and reported confidence, motivation, and focus for each participant.

	α	P1	P2	P3	P4	P5	P6	P7	P9	P10	P11	M	SD
VR-F PSA %	.88	52.0	68.8	2.1	39.6	25.0	20.8	43.8	33.3	47.9	50.0	38.3	18.8
RL-F PSA %	.89	52.0	70.8	4.2	33.3	10.4	25.0	45.8	39.6	37.5	39.6	35.8	19.4
VR-U PSA %	.91	29.1	75.0	4.2	22.9	18.8	10.4	41.7	27.1	39.6	25.0	29.4	19.7
RL-U PSA %	.96	35.4	75.0	6.3	70.8	37.5	2.1	54.2	41.7	64.6	27.1	41.5	25.2
Self-confidence	.83	3.00	4.67	4.67	4.00	3.67	4.33	3.33	2.67	2.67	3.33	3.63	0.76
Motivation	.76	4.33	4.33	4.67	5.00	4.33	4.00	2.67	4.67	5.00	3.67	4.27	0.70
Focus	.74	4.33	4.00	5.00	5.00	5.00	4.67	3.67	4.67	5.00	3.33	4.47	0.61

using VR as a presentation tool, and the perceived ability to focus attributed to the VR program, were firmly in the ‘agree’ category on average. The Kendall’s Tau correlation analysis, however, showed that among all the variables in Table 2, no significant correlations were discovered involving anxiety. Though focus did show a significant positive correlation with motivation ($\tau = .67$, $p < .05$).

In partial answer to RQ3, average fluency measures can be seen in Table 3 alongside average anxiety per condition. Results showed that articulation rates appeared little affected in VR regardless of audience, and likewise little-changed when comparing VR and RL conditions. However, the articulation rate in the RL-U condition ($Mdn = 2.26$) was significantly higher than with the RL-F audience ($Mdn = 2.18$), $z = -2.395$, $p < .05$, $r = -.54$.

With total disfluencies, only nonsignificant differences were found when comparing audience familiarity and unfamiliarity within corresponding delivery modes (VR-F & VR-U, $p = .11$; RL-F & RL-U, $p = .14$). But between VR and RL delivery modes, total disfluencies were significantly higher in the two VR conditions (VR-U $Mdn = 23$, VR-F $Mdn = 33$) compared with RL-U ($Mdn = 18$), $z = -2.601$, $p < .01$, $r = -.58$ for VR-U and $z = -2.547$, $p < .05$, $r = -.58$ for VR-F. Looking closer, repairs were found to be significantly more frequent in the RL-F condition ($Mdn = 2.86$) compared to the RL-U condition ($Mdn = 1.61$), $z = -2.803$, $p < .01$, $r = -.63$, and between the VR-F ($Mdn = 2.86$) when compared to RL-U, $z = -2.310$, $p < .05$, $r = -.53$; however, no significant differences were found among repetitions across conditions.

Also concerning RQ3, average total speaking time (TST) was compared across conditions. When

comparing familiar to unfamiliar audience conditions, a noticeable difference was found with the RL-U ($Mdn = 190$) delivery mode where participants had significantly shorter TSTs than participants to the RL-F audience ($Mdn = 203.5$), $z = -2.244$, $p < .05$, $r = -.5$ and also the VR-F audience ($Mdn = 222$), $z = -2.192$, $p < .05$, $r = -.5$. TSTs within VR were nonsignificant ($p = .77$). Relatedly, the average pruned word count, when accounting for supplied words from the bullet-point notes, showed significantly lower numbers in the RL-U condition ($Mdn = 3.08$) compared with the RL-F condition ($Mdn = 3.38$), $z = -1.989$, $p < .05$, $r = -.44$. And there was also a significant difference in pruned word count observed between the VR-F condition ($Mdn = 3.38$) and the RL-F condition ($Mdn = 3.38$), with fewer pruned words counted in VR, $z = -2.075$, $p < .05$, $r = -.48$. These values can be found in Table 4.

Finally, the frequency and average length of non-strategic silent pauses detected in the data are reported in Table 5. Length of silent pauses of one second or longer were not significantly different across all conditions, however the average frequency of these silent pauses was significantly reduced in the RL-U condition ($Mdn = 8.5$) compared to the VR-U ($Mdn = 11$), $z = -2.670$, $p < .01$, $r = -.6$, and the VR-F ($Mdn = 10$), $z = -2.521$, $p < .05$, $r = -.58$. Mean pause frequency was also significantly smaller in RL-F ($Mdn = 11.5$) compared to VR-F, $z = -2.207$, $p < .05$, $r = -.51$.

5. Discussion and implications

The purpose of this study was to explore the effects of variations in audience familiarity on speaker

Repair → 49 and the mosta interesting part of uh (0.5) is how they
50 come to an end, there are several causes uh for this
51 (2.5) the main reason um for the (0.5) situation is about
Repair → 52 violent encounters uh (1.5) uh they wi- uh their life uh
53 will be disturbed by the modern humans um (4.5) because
Repetition → 54 uh their um their life um may be um damaged by ther uh
Repetition → 55 (.) modern modern human activities (.)

Figure 3. Presentation excerpt from participant 1, VR-F condition.

Table 3. Averages for PSA, articulation rates, and disfluency measures.

	VR-F	RL-F	VR-U	RL-U
PSA	38.3%	35.8%	29.4%	41.5%
Articulation Rate in seconds (SD)	2.08 (0.34)	2.14 (0.32)	2.18 (0.31)	2.29 (0.32)
Disfluencies				
Repetitions per minute (SD)	5.39 (1.63)	3.49 (1.95)	4.19 (2.22)	3.81 (1.98)
Repairs per minute (SD)	3.17 (1.71)	3.02 (1.44)	2.79 (1.68)	1.73 (0.95)
Total per minute (SD)	8.57 (0.94)	6.51 (2.1)	6.98 (1.93)	5.55 (2.08)

Table 4. Pruned word count and total speaking time.

	VR-F	RL-F	VR-U	RL-U
Average pruned word count divided by supplied words	3.56	3.74	3.52	3.07
Average total speaking time in seconds (SD)	240.4 (97.5)	233.6 (108.7)	243.8 (113.4)	185.6 (39.8)

anxiety and performance between VR and RL presentation delivery modes. In summary, PSA was little changed between VR and RL when the audience remained familiar, however, an audience of unfamiliar, computer-generated agents (VR-U) elicited significantly less anxiety than the virtual audience with photorealistic agents of people familiar to the speakers (VR-F). Furthermore, presenting to a RL audience of strangers was found to produce the most anxiety, and significantly more than was detected in the VR-U condition.

Fluency did not correlate with anxiety, but alongside other performance features showed some significant differences between conditions. The presentations to the RL-U audience of strangers were marked by faster articulation, fewer repairs, shorter speaking times, fewer words spoken, and fewer silent pauses. Yet differences in these performance features were not detected between VR audiences.

Individual differences in motivation, self-confidence, and focus did not correlate with anxiety or performance.

It therefore seems that variations in virtual audiences, such as the familiarity of the audience, can impact

Table 5. Mean length, standard deviation, and mean frequency of silent pauses of one second or longer.

	VR-F	RL-F	VR-U	RL-U
Mean pause length (seconds)	2.95	2.61	2.46	2.49
SD of pause length	0.75	0.85	0.71	1.54
Mean pause frequency	17.11	12.4	14.8	8.9

speaker anxiety, though this may not affect performance. However, it was also noticed that speaking performance was affected between presenting in VR and presenting in RL.

5.1. Some variations in VR audience appearance may alleviate PSA

RQ1 inquired how audience variations may have associated with reported PSA in both RL and VR conditions. PSA with the VR-F audience was significantly greater than with the VR-U audience. This result was surprising, as studies have shown that a familiar audience usually causes less anxiety (MacIntyre and Thivierge 1995). Perhaps the less realistic rendering of the VR-U audience allowed participants to feel that the audience were not in fact real, which lessened anxiety, as opposed to the scanned 3D images of familiar faces. However, research in social presence has yet to show if photorealism is an antecedent to a sense of being co-located with someone in a virtual environment (Oh, Bailenson, and Welch 2018). Furthermore, a study by Felnhofer et al. (2019) found no significant association between social presence and anxiety in individuals without preexisting social anxiety disorders. Therefore, it seems unlikely that participants' anxiety levels with the familiar virtual audiences were influenced by the fidelity of the graphics.

When presenting in front of an unfamiliar audience in RL, participants experienced the most PSA, and significantly more than with the VR-U audience. Although the high PSA in RL-U was not unexpected since presenting to strangers in RL is an established cause of PSA (Menzel and Carrell 1994), the difference in PSA between the RL-U and VR-U condition demonstrates that VR can be a useful tool for practicing or delivering speeches for individuals who are prone to PSA, as certain virtual audiences may mitigate PSA.

5.2. Individual differences in motivation, focus, and self-confidence may have no effect on PSA or performance, or may have an indirect effect

RQ2 considered the effects of speaker individual differences on PSA and fluency in presentations. Although no correlations were detected, the descriptive statistics show that on average participants were motivated by the VR presentation tool and that it also helped them to focus on the presentation task. The inferential results could indicate that these factors do not directly correlate with PSA, and instead are associated in some other way, but in all likelihood, either the low number of participants in this study or an insensitivity of the instrument did not allow for meaningful results with this analysis.

5.3. Performance was improved in high PSA conditions, and some features of performance were more prominent in VR than in RL

RQ3 was concerned with the association between reported PSA and speaking performance. Fluency was found to be marked in the RL-U condition compared to others. Articulation rates were higher and repairs less frequent when participants presented to an unfamiliar audience in RL than to a familiar one. Furthermore, in the RL conditions, speakers spoke less in terms of both time and pruned-word count when confronted with an unfamiliar audience compared to a familiar one. Lastly, non-strategic silent pauses were much less frequent in RL-U compared to both VR conditions.

It is known that anxiety can negatively affect word count in presentations (Castillejo 2018; Hewitt and Stephenson 2012), which explains the lower pruned-word count in the high-anxiety RL-U condition, and coupled with faster articulation, also explains the shorter presentation times in this condition. Unexpectedly though, disfluencies, especially repairs, decreased in the RL-U condition, and silent pauses were also remarkably reduced. The relative decrease in repairs and silent pauses in RL-U may be attributed to an increase in self-monitoring when confronted with a RL audience of strangers. This could perhaps be from a facilitative level of anxiety or an impulse to avoid losing face in front of strangers, but more research is needed to see if this can be replicated and what may be causing it.

Also of interest, participants tended to demonstrate specific performance features more in VR than in RL. For example, non-strategic silent pauses were significantly more frequent in VR conditions than in RL, total speaking times were uniformly longer in VR, and total disfluencies were higher in VR. A portion of this can be attributed to a repetition affect, wherein the second iteration of a presentation is slightly more fluent, but an additional explanation can be that speakers naturally self-monitored less when presenting in VR and may not have felt the awkwardness that silent pauses can produce when they occur in front of live audiences.

5.4. Implications

Monologic speaking tasks such as presentations, speeches, and oral proficiency interviews are commonly conducted in language learning contexts for purposes of assessing speaking. The results reported in this study inform teachers of how variations in the instructional environment of presentation-style assessments will affect anxiety and performance.

It was confirmed that presenting to an audience of strangers in RL causes higher levels of PSA, reducing total speaking time, however, it was discovered that this format also seems to improve some features of performance. Reduction in PSA can be achieved through VR, but VR can also negatively impact some speaking performance features. Teachers may be able to mitigate this and improve performance in VR presentations by telling students they are being assessed, which may increase speakers' self-monitoring. Furthermore, using VR for presentation assessments may prove a useful method to control the instructional environment to provide for more breadth and depth of speaking available for assessment. Speakers who are predisposed to debilitating anxiety may find that VR provides an environment that is conducive, or even necessary, to their success in speaking assessments.

Educators who reject the thought of accepting VR as a venue for public speaking might instead use it for exposure therapy in a scaffolded instructional design. VR could provide low-anxiety scenarios for speakers to practice in before moving on to higher-anxiety scenarios and ultimately to RL presentations.

6. Limitations and contributions

Due to this study's limitations the results should be considered as tentative. Although exploratory studies in human computer interaction commonly involve fewer than 15 participants (Caine 2016), a larger dataset would have provided more robust measurements. An *a priori* power analysis for a one-tailed Wilcoxon signed rank test using G*Power, given $p = .8$ and α set to .05, recommends a sample of $n = 28$ (Faul et al. 2007). Another limitation was the possibility of an order effect where participants' fluency may have improved in the second iteration of the presentation. Although any change in fluency is likely to have been negligible (Baker, Laurence, and Mondloch 2017), the authors recommend avoiding this repetition in similar studies to leave less doubt about results. Furthermore, a possible validity threat exists for the group that was randomly assigned to the unfamiliar conditions first, as they had more exposure to the researchers than the other groups before presenting in the familiar conditions and thus may have developed a higher degree of familiarity.

This research provides important insight into the effect of audience familiarity on spoken performance in a monologic speaking task delivered in virtual reality and real-life modalities. A primary contribution of this study was to show that presentations to virtual audiences were marked by more disfluencies than presentations to real audiences. Furthermore, this study

confirmed that PSA is observed in virtual reality, but that audience familiarity may impact PSA.

7. Conclusion and future directions

Results from this study indicated that variations in virtual audiences may impact public speaking anxiety (PSA). Virtual audiences that are similar in appearance to real people known to the participants appeared to produce more PSA than a virtual audience of computer-generated agents. Additionally, presentation performance was impacted differently when presenting in VR and when presenting to an audience of strangers in RL. VR can be a useful tool for manipulating the instructional environment to optimise speaker performance. Institutions should consider adopting VR as education continues to evolve to become more personalised and blends boundaries between the virtual and the actual.

This exploratory study could be expanded in many directions by, for example, examining additional performance measures (e.g. body or eye tracking), different tasks (e.g. spontaneous speech on familiar topics, or IELTS-style proficiency interviews), and different audience types (e.g. gender or age of audience) to see if these variables produce similar or different results.

Notes

1. <https://www.ovationvr.com/>.
2. <https://virtualspeech.com/>.

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Ethics statement

This research was approved by the university's ethics committee and followed all required procedures to ensure the safety and consent of the participants.

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References

- Baker, K. A., S. Laurence, and C. J. Mondloch. 2017. "How Does a Newly Encountered Face Become Familiar? The Effect of Within-Person Variability on Adults' and Children's Perception of Identity." *Cognition* 161: 19–30. doi:10.1016/j.cognition.2016.12.012.
- Bandura, A. 1997. *Self-Efficacy: The Exercise of Control*. New York: W. H. Freeman and Company.
- Bankowski, E. 2010. "Developing Skills for Effective Academic Presentations in EAP." *International Journal of Teaching and Learning in Higher Education* 22 (2): 187–196.
- Barreda-Ángeles, M., S. Alex-Guillaume, and A. Pareda-Baños. 2020. "Users' Psychophysiological, Vocal, and Self-Reported Responses to the Apparent Attitude of a Virtual Audience in Stereoscopic 360°-Video." *Virtual Reality* 24: 289–302. doi:10.1007/s10055-019-00400-1.
- Boetje, J., and S. van Ginkel. 2021. "The Added Benefit of an Extra Practice Session in Virtual Reality on the Development of Presentation Skills: A Randomized Control Trial." *Journal of Computer Assisted Learning* 37: 253–264. doi:10.1111/jcal.12484.
- Bouchard, S., J. Bossé, C. Loranger, and É. Klinger. 2014. "Social Anxiety Disorder: Efficacy and Virtual Humans." In *Advances in Virtual Reality and Anxiety Disorders*, edited by B. Wiederhold and S. Bouchard, 187–209. doi:10.1007/978-1-4899-8023-6_9.
- Caine, K. 2016. "Local Standards for Sample Size at CHI." Paper Presented at the 2016 CHI Conference on Human Factors in Computing Systems. Academic Press. doi:10.1145/2858036.2858498.
- Carlin, A., H. Hoffman, and S. Weghorst. 1997. "Virtual Reality and Tactile Augmentation in the Treatment of Spider Phobia: A Case Report." *Behaviour Research and Therapy* 35 (2): 153–158. doi:10.1016/s0005-7967(96)00085-x.
- Castillejo, S. P. 2018. "The Role of Foreign Language Anxiety on L2 Utterance Fluency During a Final Exam." *Language Testing* 1 (19): 1–19. doi:10.1177/0265532218777783.
- Chen, J. 2020. "The Effects of Pre-Task Planning on EFL Learners' Oral Performance in 3D Multi-User Virtual Environments." *ReCALL First View*, 1–18. doi:10.1017/S0958344020000026.
- Chollet, M., and S. Scherer. 2017. "Perception of Virtual Audiences." *IEEE Computer Graphics and Applications* 37 (4): 50–59. doi:10.1109/MCG.2017.3271465.
- Clément, R., Z. Dörnyei, and K. Noels. 1994. "Motivation, Self-Confidence, and Group Cohesion in the Foreign Language Classroom." *Language Learning* 44 (3): 417–448. doi:10.1111/j.1467-1770.1994.tb01113.x.
- Council of Europe. 2001. *Common European Framework of Reference for Languages: Learning, Teaching, Assessment*. Press Syndicate of the University of Cambridge.
- Dalgarno, B., and M. Lee. 2010. "What are the Learning Affordances of 3-D Virtual Environments?" *British*

- Journal of Educational Technology* 41 (1): 10–32. doi:10.1111/j.1467-8535.2009.01038.x.
- Daly, J., A. Vangelisti, and S. Lawrence. 1989. "Self-Focused Attention and Public Speaking Anxiety." *Personality and Individual Differences* 10 (8): 903–913. doi:10.1016/0191-8869(89)90025-1.
- Davis, A., D. L. Linvill, L. F. Hodges, A. F. Da Costa, and A. Lee. 2019. "Virtual Reality Versus Face-to-Face Practice: A Study into Situational Apprehension and Performance." *Communication Education*, 1–15. doi:10.1080/03634523.2019.1684535.
- De Jong, N. H. 2018. "Fluency in Second Language Testing: Insights from Different Disciplines." *Language Assessment Quarterly* 15 (3): 237–254. doi:10.1080/15434303.2018.1477780.
- Dörnyei, Z., and T. Taguchi. 2010. *Questionnaires in Second Language Acquisition: Construction, Administration, and Processing*. 2nd ed. New York: Routledge.
- Dubiago, M., S. Poeschl, and N. Doering. 2017. "Influence of Simulation Fidelity on Perceived Simulation Realism: An Exploratory Study on a Virtual Public Speaking Training Application." In *Annual Review of CyberTherapy and Telemedicine*, edited by B. Wiederhold, G. Riva, C. Fullwood, A. Attrill-Smith, and G. Kirwan, 154–160. San Diego: Interactive Media Institute.
- Faul, F., E. Erdfelder, A.-G. Lang, and A. Buchner. 2007. "G*Power 3: A Flexible Statistical Power Analysis for the Social, Behavioral, and Biomedical Sciences." *Behavior Research Methods* 39: 175–191. doi:10.3758/BF03193146.
- Felinhofer, A., H. Hlavacs, L. Beutl, I. Kryspin-Exner, and O. D. Kothgassner. 2019. "Physical Presence, Social Presence, and Anxiety in Participants with Social Anxiety Disorder During Virtual Cue Exposure." *Cyberpsychology, Behavior, and Social Networking* 22 (1): 46–50. doi:10.1089/cyber.2018.0221.
- Fernandes, L. M. A., G. C. Matos, D. Azevedo, R. R. Nunes, H. Paredes, L. Morgado, L. F. Barbosa, et al. 2016. "Exploring Educational Immersive Videogames: An Empirical Study with a 3D Multimodal Interaction Prototype." *Behaviour & Information Technology* 35 (11): 907–918. doi:10.1080/0144929x.2016.1232754.
- Field, A. 2009. *Discovering Statistics Using SPSS*. 3rd ed. Los Angeles: Sage.
- Ghanbarpour, M. 2016. "Willingness to Communicate, Linguistic Self-Confidence, and Language-Use Anxiety: The Iranian EFL Context." *Theory and Practice in Language Studies* 6 (12): 2265–2271. doi:10.17507/tpls.0612.05.
- Gilkinson, H. 1942. "Social Fears as Reported by Students in College Speech Classes." *Speech Monographs* 9 (1): 141–160. doi:10.1080/03637754209390068.
- Glémarec, Y., J.-C. Lugin, A.-G. Bosser, A. C. Jackson, C. Buche, and M. E. Latoschik. 2021. "Indifferent or Enthusiastic? Virtual Audiences Animation and Perception in Virtual Reality." *Frontiers in Virtual Reality*, 1–16. doi:10.3389/frvir.2021.666232.
- Gruber, A., and R. Kaplan-Rakowski. 2020. "User Experience of Public Speaking Practice in Virtual Reality." In *Cognitive and Affective Perspectives on Immersive Technology in Education*, edited by R. Zheng, 235–249. IGI-Global. doi:10.4018/978-1-7998-3250-8.
- Harris, S., R. Kemmerling, and M. North. 2002. "Brief Virtual Reality Therapy for Public Speaking Anxiety." *CyberPsychology & Behavior* 5 (6): 543–550. doi:10.1089/109493102321018187.
- Heuett, B., and K. Heuett. 2011. "Virtual Reality Therapy: A Means of Reducing Public Speaking Anxiety." *International Journal of Humanities and Social Science* 1 (16): 1–6.
- Hewitt, E., and J. Stephenson. 2012. "Foreign Language Anxiety and Oral Exam Performance: A Replication of Phillips's MLJ Study." *The Modern Language Journal* 96 (2): 170–189. doi:10.1111/j.1540-4781.2011.01174.x.
- Hook, J. N., C. A. Smith, and D. P. Valentiner. 2008. "A Short-Form of the Personal Report of Confidence as a Speaker." *Personality and Individual Differences* 44: 1306–1313. doi:10.1016/j.paid.2007.11.021.
- Kang, N., W.-P. Brinkman, M. B. van Riemsdijk, and M. Neerincx. 2014. "An Expressive Virtual Audience with Flexible Behavioral Styles." *IEEE Transactions on Affective Computing*. doi:10.1109/TAFFC.2013.2297104.
- Kang, N., W.-P. Brinkman, M. B. van Riemsdijk, and M. Neerincx. 2016. "The Design of Virtual Audiences: Noticeable and Recognizable Behavioral Styles." *Computers in Human Behavior* 55: 680–694. doi:10.1016/j.chb.2015.10.008.
- Kazemi, S. A., and L. Zarei. 2015. "The Efficacy of Topic Familiarity on Oral Presentation: Extensive Speaking Assessment Task of Iranian EFL Learners in TBLT." *International Journal of Applied Linguistics & English Literature* 4 (3): 93–97. doi:10.7575/aiac.ijalel.v.4n.3p.93.
- King, P. E., and A. N. Finn. 2017. "A Test of Attention Control Theory in Public Speaking: Cognitive Load Influences the Relationship Between State Anxiety and Verbal Production." *Communication Education* 66 (2): 168–182. doi:10.1080/03634523.2016.1272128.
- Lan, Y.-J. 2020. "Immersion, Interaction and Experience-Oriented Learning: Bringing Virtual Reality into FL Learning." *Language Learning & Technology* 24 (1): 1–15.
- Lin, T.-J., and Y.-J. Lan. 2015. "Language Learning in Virtual Reality Environments: Past, Present, and Future." *Educational Technology & Society* 18 (4): 486–497.
- Lister, H., C. D. Piercey, and C. Joordens. 2010. "The Effectiveness of 3-D Video Virtual Reality for the Treatment of Fear of Public Speaking." *Journal of CyberTherapy & Rehabilitation* 3 (4): 375–381.
- MacIntyre, P. D. 2017. "An Overview of Language Anxiety Research and Trends in Its Development." In *New Insights into Language Anxiety: Theory, Research and Educational Implications*, edited by C. Gkonou, M. Daubney, and J. M. Dewaele, 11–30. Multilingual Matters. doi:10.21832/9781783097722-003.
- MacIntyre, P. D., and K. A. Thivierge. 1995. "The Effects of Audience Pleasantness, Audience Familiarity, and Speaking Contexts on Public Speaking Anxiety and Willingness to Speak." *Communication Quarterly* 43 (4): 456–466. doi:10.1080/01463379509369992.
- Matsuda, S., and P. Gobel. 2004. "Anxiety and Predictors of Performance in the Foreign Language Classroom." *System* 32: 21–36. doi:10.1016/j.system.2003.08.002.

- Menzel, K., and L. Carrell. 1994. "The Relationship Between Preparation and Performance in Public Speaking." *Communication Education* 43 (1): 17–26. doi:10.1080/03634529409378958.
- Mostajeran, F., M. B. Balci, F. Steinicke, S. Kühn, and J. Gallinat. 2020. "The Effects of Virtual Audience Size on Social Anxiety During Public Speaking." In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 303–312. doi:10.1109/VR46266.2020.00050.
- Nazligul, M. D., M. Yilmaz, U. Gulec, A. E. Yilmaz, V. Isler, R. V. O'connor, M. A. Gozcu, and P. Clarke. 2018. "Interactive Three-Dimensional Virtual Environment to Reduce the Public Speaking Anxiety Levels of Novice Software Engineers." *IET Software* 13 (2): 152–158. doi:10.1049/iet-sen.2018.5140.
- Oh, C. S., J. N. Bailenson, and G. F. Welch. 2018. "A Systematic Review of Social Presence: Definition, Antecedents, and Implications." *Frontiers in Robotics and AI* 5 (114). doi:10.3389/frobt.2018.00114.
- Piniel, K., and K. Csizér. 2015. "Changes in Motivation, Anxiety and Self-Efficacy During the Course of an Academic Writing Seminar." In *Motivational Dynamics in Language Learning*, edited by Z. Dörnyei, P. D. MacIntyre, and A. Henry, 164–194. Bristol: Multilingual Matters.
- Raja, F. 2017. "Anxiety Level in Students of Public Speaking: Causes and Remedies." *Journal of Education and Educational Development* 4 (1): 94–110. doi:10.22555/joeed.v4i1.1001.
- Schwiehorst, K. 2002. "The State of VR: A Meta-Analysis of Virtual Reality Tools in Second Language Acquisition." *Computer Assisted Language Learning* 15 (3): 221–239. doi:10.1076/call.15.3.221.8186.
- Segalowitz, N. 2010. *Cognitive Bases of Second Language Fluency*. New York: Taylor & Francis.
- Sharmili, A., and R. Kanagaraj. 2020. "Live Beyond Fear: A Virtual Reality Serious Game Platform to Overcome Phobias." In *IEEE 5th International Conference on Devices, Circuits and Systems (ICDCS)*, 336–339. doi:10.1109/ICDCS48716.2020.243592.
- Sherman, W., and A. Craig. 2019. *Understanding Virtual Reality: Interface, Application, and Design*. 2nd ed.. Cambridge, US: Morgan Kaufmann.
- Slater, M. 2009. "Place Illusion and Plausibility Can Lead to Realistic Behavior in Immersive Virtual Environments." *Philosophical Transactions of the Royal Society B* 364: 3549–3557. doi:10.1098/rstb.2009.0138.
- Slater, M., D.-B. Pertaub, C. Barker, and D. Clark. 2006. "An Experimental Study on Fear of Public Speaking Using a Virtual Environment." *Cyberpsychology & Behavior* 9 (5): 627–633. doi:10.1089/cpb.2006.9.627.
- Slater, M., and M. Sanches-Vives. 2016. "Enhancing Our Lives with Immersive Virtual Reality." *Frontiers in Robotics and AI* 3 (74): 1–47. doi:10.3389/frobt.2016.00074.
- Spielberger, C., ed. 1966. *Anxiety and Behavior*. London: Academic Press.
- Stupar-Rutenfrans, S., L. Ketelaars, and M. van Gisbergen. 2017. "Beat the Fear of Public Speaking: Mobile 360° Video Virtual Reality Exposure Training in Home Environment Reduces Public Speaking Anxiety." *Cyberpsychology, Behavior, and Social Networking* 20 (10): 624–633. doi:10.1089/cyber.2017.0174.
- Sun, B., and A. Révész. 2021. "The Effects of Task Repetition on Child EFL Learners' Oral Performance." *Canadian Journal of Applied Linguistics* 24 (2): 30–47. doi:10.37213/cjal.2021.31382.
- Takac, M., J. Collett, K. J. Blom, R. Conduit, I. Rhem, and A. De Foe. 2019. "Public Speaking Anxiety Decreases Within Repeated Virtual Reality Training Sessions." *PLoS One* 14 (5): 1–17. doi:10.1371/journal.pone.0216288.
- Tudor, A.-D., S. Poeschl, and N. Doering. 2013. "Virtual Audience Customization for Public Speaking Training Procedures." *IEEE Virtual Reality*, 61–62. doi:10.1109/VR.2013.6549363.
- Wilsdon, L., and C. Fullwood. 2017. "The Effect of Immersion and Presence in a Virtual Reality Public Speaking Task." In *Annual Review of CyberTherapy and Telemedicine*, edited by B. Wiederhold, G. Riva, C. Fullwood, A. Attrill-Smith, and G. Kirwan, 211–213. San Diego: Interactive Media Institute.
- Woodrow, L. 2006. "Anxiety and Speaking English as a Second Language." *Regional Language Centre Journal* 37 (3): 308–328. doi:10.5539/ass.v10n12p170.
- Wörtwein, T., L.-P. Morency, and S. Scherer. 2015. "Automatic Assessment and Analysis of Public Speaking Anxiety: A Virtual Audience Case Study. In *The Proceedings of the International Conference on Affective Computing and Intelligent Interaction (ACII)*, 187–193. doi:10.1109/ACII.2015.7344570.
- Yan, J. X., and E. K. Horwitz. 2008. "Learners' Perceptions of How Anxiety Interacts with Personal and Instructional Factors to Influence Their Achievement in English: A Qualitative Analysis of EFL Learners in China." *Language Learning* 58 (1): 151–183. doi:10.1111/j.1467-9922.2007.00437.x.
- Yoshimura, Y., and B. MacWhinney. 2007. "The Effect of Oral Repetition on L2 Speech Fluency: An Experimental Tool and Language Tutor." In *The Proceedings of Speech and Language Technology in Education*, 25–28.
- Zhou, H., Y. Fujimoto, M. Kanbara, and H. Kato. 2021. "Virtual Reality as a Reflection Technique for Public Speaking Training." *Applied Sciences* 11 (3988): 1–18. doi:10.3390/app11093988.